

# Evolutionary Individuality as an Empirical Question

## *A Proposed Experimental Test in Pseudomonas*

### Introduction

Major evolutionary transitions raise a deceptively difficult question: when does a collection of lower-level entities become an evolutionary individual in its own right? The problem is not merely whether selection can occur among collectives. Different accounts of evolutionary individuality place different weight on the processes that make higher-level selection possible, on whether the organization supporting those processes is internally carried or environmentally supplied, and on whether the collective participates in a reproducible life cycle of its own.<sup>1</sup> These differences matter most in systems near a transition, where a collective can display substantial higher-level organization without satisfying every proposed criterion for individuality.

Ecological scaffolding makes this problem experimentally accessible. Environmental structure can organize lower-level entities into collectives that display higher-level variation, differential reproductive success, and heredity even when some of the causal organization supporting those properties remains external to the collective itself.<sup>2</sup> Such systems create an important distinction between the appearance or operation of higher-level Darwinian properties and the question of whether the collective has come to carry the organization that sustains them. Bourrat's endogenization account makes that distinction explicit: externally generated collective organization can support a Darwinian-like state, whereas stronger evolutionary individuality requires the relevant individuating organization to become internally generated or sufficiently carried by the collective.<sup>3</sup>

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<sup>1</sup> Ellen Clarke, "A Levels-of-Selection Approach to Evolutionary Individuality," *Biology & Philosophy* 31 (2016): 893–911, <https://doi.org/10.1007/s10539-016-9540-4>; "Evolutionary Individuals," in *The Units of Life: Kinds of Individual in Biology* (Oxford University Press, 2025), 81–120, <https://doi.org/10.1093/9780191948008.003.0005>; Pierrick Bourrat, "Evolutionary Transitions in Individuality by Endogenization of Scaffolded Properties," *The British Journal for the Philosophy of Science* 76, no. 2 (2025): 269–294, <https://doi.org/10.1086/719118>; Guilhem Doulier, Peter Takacs, and Pierrick Bourrat, "Evolutionary Transitions in Individuality and Life Cycle Closure," *Philosophy of Science* 91, no. 5 (2024): 1393–1403, <https://doi.org/10.1017/psa.2023.162>.

<sup>2</sup> Andrew J. Black, Pierrick Bourrat, and Paul B. Rainey, "Ecological Scaffolding and the Evolution of Individuality," *Nature Ecology & Evolution* 4 (2020): 426–436, <https://doi.org/10.1038/s41559-019-1086-9>.

<sup>3</sup> Pierrick Bourrat, "Evolutionary Transitions in Individuality by Endogenization of Scaffolded Properties," *The British Journal for the Philosophy of Science* 76, no. 2 (2025): 269–294, <https://doi.org/10.1086/719118>.

Other accounts place the emphasis elsewhere. On Clarke's levels-of-selection approach, evolutionary individuality concerns a collective's capacity to respond to selection at its own level, grounded by objective individuating mechanisms such as policing or demarcation.<sup>4</sup> On the life-cycle account developed by Doulicier, Takacs, and Bourrat, the central question is whether a collective-level life cycle exhibits multiplication and closure among biologically meaningful stages.<sup>5</sup> These accounts overlap, but they are not interchangeable. A collective could in principle possess a genuine higher-level selection capacity and a closed collective life cycle while still depending on an external ecological relation for part of the organization that makes those properties possible. Conversely, loss of one proposed criterion need not imply that every other form of higher-level organization has disappeared.

This creates an empirical opportunity. Rather than asking which account supplies the single correct definition of evolutionary individuality, one can ask what the same biological evidence warrants under several serious accounts whose commitments are kept distinct. The most informative case is not a mature organism in which the major features of individuality already coincide, but a transitional system in which externally structured and internally carried organization can be experimentally separated.

The proposed study uses an experimentally evolved *Pseudomonas fluorescens* system to make that separation testable. The central question is temporal as well as causal: after a collective-lineage organization has evolved under an ecology that protects lineage-level structure, what happens when part of that external protection is subsequently relaxed? If the evolved collective retains the relevant organization, that would support some degree of independence from that scaffold relation. If the organization proves dependent on the protected ecology, the same collective may remain a higher-level evolutionary entity under some accounts while falling short of stronger endogenization requirements under another. Either result would be informative because the experiment is designed to preserve the biological target while allowing the accounts to give different answers.

## **An Experimental System for a Post-Evolution Scaffold Challenge**

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<sup>4</sup> Ellen Clarke, "A Levels-of-Selection Approach to Evolutionary Individuality," *Biology & Philosophy* 31 (2016): 893–911, <https://doi.org/10.1007/s10539-016-9540-4>; "Evolutionary Individuals," in *The Units of Life: Kinds of Individual in Biology* (Oxford University Press, 2025), 81–120, <https://doi.org/10.1093/9780191948008.003.0005>.

<sup>5</sup> Guilhem Doulicier, Peter Takacs, and Pierrick Bourrat, "Evolutionary Transitions in Individuality and Life Cycle Closure," *Philosophy of Science* 91, no. 5 (2024): 1393–1403, <https://doi.org/10.1017/psa.2023.162>.

The experimental evolution of *Pseudomonas fluorescens* provides an unusually well-developed system for posing this question. In the lineage of experiments initiated by Hammerschmidt and colleagues, bacterial collectives were propagated under selection favoring collective persistence through an alternating phenotypic life cycle involving wrinkly spreader (WS) and smooth (SM) states.<sup>6</sup> In the resulting two-phase cycle, WS cells form mat-like collectives associated with a soma-like phase, whereas SM cells serve a propagule-like or germ-like role capable of founding a subsequent WS stage. Selection under the cheat-embracing regime produced lineages in which reliable transition between these phases became increasingly important to collective persistence, and improvements in lineage fitness could become decoupled from changes in the fitness of constituent cells.<sup>7</sup>

This experimental architecture was later used to examine how metapopulation structure alters the balance of selection. Rose and colleagues compared two ecologies built around the same broad WS–SM–WS life cycle.<sup>8</sup> In the Non-Mixed Propagule treatment, propagules from different surviving collective lineages remained separated during the dispersal phase. In the Mixed Propagule treatment, propagules from surviving groups were pooled before the next stage of the life cycle. New mats in both treatments were founded through a single-cell bottleneck, but the dispersal environments differed in whether propagule lineages remained protected from direct cross-lineage competition.<sup>9</sup>

That difference changed the evolutionary regime. Under the Non-Mixed treatment, competition was concentrated more strongly among collective lineages. Under the Mixed treatment, direct competition among propagule cells intensified, reducing the opportunity associated with between-lineage variation and shifting evolutionary response toward traits that improved lower-level cell performance rather than traits associated with collective life-cycle performance.<sup>10</sup> The comparison therefore provides more than two environmental treatments: it identifies a specific ecological relation—protected propagule-lineage structure—that can contribute to maintaining higher-level evolutionary differentiation.

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<sup>6</sup> Katrin Hammerschmidt, Caroline J. Rose, Benjamin Kerr, and Paul B. Rainey, “Life Cycles, Fitness Decoupling and the Evolution of Multicellularity,” *Nature* 515 (2014): 75–79, <https://doi.org/10.1038/nature13884>.

<sup>7</sup> Ibid.

<sup>8</sup> Caroline J. Rose, Katrin Hammerschmidt, Yuriy Pichugin, and Paul B. Rainey, “Meta-population Structure and the Evolutionary Transition to Multicellularity,” *Ecology Letters* 23 (2020): 1380–1390, <https://doi.org/10.1111/ele.13570>.

<sup>9</sup> Ibid.

<sup>10</sup> Ibid.

Subsequent work has strengthened the biological interpretation of the system. Doulcier and colleagues reconstructed genealogies of nascent multicellular lineages and showed that reliable switching between soma-like and germ-like phases can make a substantial contribution to lineage persistence.<sup>11</sup> Their experiments also document failure modes at the lineage level, including failure to generate the required next-stage cells, collapse of the mat phase, and lineage extinction. These features make the system particularly useful for a proposed test of individuality because success and failure can both be represented within the same collective-lineage history rather than inferred only from surviving lineages.

The candidate biological individual in the proposed study is therefore not the culture vessel, the WS mat considered in isolation, the SM propagule pool, or the entire metapopulation. It is the genealogically traceable collective lineage spanning the WS–SM–WS life cycle. The metapopulation is the level at which variation, reproductive contribution, heredity, and selection among such lineages can be measured. This distinction is important because the experimental treatment acts at the level of metapopulation structure, whereas the candidate individual whose status is being evaluated is the collective lineage.

The experiments reviewed here establish that protected versus mixed propagule structure can alter evolutionary response, but they do not answer the specific post-evolution question proposed here. Rose and colleagues allowed lineages to evolve under the two ecological regimes from the outset.<sup>12</sup> The present proposal instead asks what happens after collective-lineage organization has first evolved under the protected Non-Mixed ecology and matched derived material is subsequently exposed either to continued lineage protection or to relaxed propagule-lineage structure. That change in temporal order is essential. It converts a comparison of evolutionary environments into a test of what an already evolved collective organization continues to carry when one identified part of its ecological scaffold is altered.<sup>13</sup>

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<sup>11</sup> Guilhem Doulcier, Philippe Remigi, Daniel Rexin, and Paul B. Rainey, “Evolutionary Dynamics of Nascent Multicellular Lineages,” *Proceedings of the Royal Society B* 292 (2025): 20241195, <https://doi.org/10.1098/rspb.2024.1195>.

<sup>12</sup> Caroline J. Rose, Katrin Hammerschmidt, Yuriy Pichugin, and Paul B. Rainey, “Meta-population Structure and the Evolutionary Transition to Multicellularity,” *Ecology Letters* 23 (2020): 1380–1390, <https://doi.org/10.1111/ele.13570>.

<sup>13</sup> Guilhem Doulcier, Peter Takacs, Katrin Hammerschmidt, et al., “Stability of Ecologically Scaffolded Traits during Evolutionary Transitions in Individuality,” *Nature Communications* 15 (2024): 6566, <https://doi.org/10.1038/s41467-024-50625-1>.

For that reason, the preferred reference material would consist of independently evolved endpoint backgrounds with documented Non-Mixed evolutionary provenance, selected because their history and biological material are suitable for the test rather than because they already support a desired assessment of individuality. Prior experimental results would establish feasibility and guide the design, but they would not substitute for prospective characterization of the reference state. If suitable independently evolved material is not available, the proposal would require a narrower reconstruction strategy and correspondingly narrower claims.

The primary contrast would then preserve the protected Non-Mixed propagule-lineage architecture in one branch while relaxing that relation through the Mixed Propagule ecology in the other. The proposed causal claim is intentionally limited: this comparison tests dependence on the protected propagule-lineage relation as a whole. It does not by itself distinguish the consequences of reduced lineage assortment from those of increased cross-lineage competition, because both arise when propagules are pooled. A more finely decomposed mechanism test would require an additional experiment.

The same caution applies to any observed decline in collective performance after scaffold relaxation. Poorer performance in the Mixed ecology would not by itself show that the evolved collective had lost internally carried organization; the altered ecology could simply make an otherwise retained collective trait less advantageous or less strongly expressed. The proposed study therefore pairs measurements made within the challenge environment with a standardized common-environment re-assay of descendants with established genealogical links to the starting lineages. The purpose is to distinguish immediate ecological performance from heritable retention, erosion, or reorganization of the collective traits that evolved under the protected ecology.

This post-evolution challenge creates an experimental setting in which the three accounts become separable. Clarke's account asks whether the collective retains a genuine higher-level selection capacity grounded by an objective individuating mechanism. Bourrat's account asks whether the relevant Darwinian organization is externally scaffolded or has become internally carried. The life-cycle account asks whether collective multiplication and closure remain supported for the same lineage-level entity. The proposed experiment does not assume that these questions

will have the same answer. Its purpose is to make their agreement or disagreement empirically visible while keeping the lineage under study unchanged.

### Three Accounts of the Same Evolving Collective

The proposed experiment is useful only if the accounts being compared retain their own criteria. A difference in assessment would be uninformative if it arose because one account had been translated into the terms of another. The relevant question is therefore not whether Clarke's levels-of-selection account, Bourrat's endogenization account, and the life-cycle-closure account can be reduced to a common definition of individuality. It is whether the same experimentally characterized collective lineage can satisfy the empirical criteria of each account separately.

Clarke's account begins from the role of individuals as units capable of evolution by natural selection. In its current formulation, individuating mechanisms—including policing and demarcating mechanisms—determine whether a compositional unit has the relevant evolutionary capacity: policing mechanisms constrain selection among parts of a candidate individual, whereas demarcating mechanisms maintain or increase the conditions for selection among candidate individuals.<sup>14</sup> The account is deliberately functional. What matters is the contribution of a mechanism to the relevant selection capacity rather than a requirement that individuality always be realized through a particular material device such as a specific bottleneck, germ–soma distinction, or physical boundary. Clarke also distinguishes historical from forwards-looking ways of applying the concept; the proposed experiment adopts the forwards-looking question of what the evolved collective is capable of under the conditions in which it is tested.<sup>15</sup>

That emphasis matters in the *Pseudomonas* system because protected propagule-lineage structure is a plausible candidate demarcating mechanism. Keeping propagules from different collective lineages separate preserves differences among lineages during the dispersal phase, whereas pooling those propagules exposes them to direct competition before the next collective generation is established. The proposed study does not assume that this ecological relation is an individuating mechanism merely because it was imposed experimentally. Its status would depend on whether the relation can be

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<sup>14</sup> Ellen Clarke, “A Levels-of-Selection Approach to Evolutionary Individuality,” *Biology & Philosophy* 31 (2016): 893–911, <https://doi.org/10.1007/s10539-016-9540-4>; “Evolutionary Individuals,” in *The Units of Life: Kinds of Individual in Biology* (Oxford University Press, 2025), 81–120, <https://doi.org/10.1093/9780191948008.003.0005>.

<sup>15</sup> Ibid.

shown to contribute causally to the collective lineage's capacity for between-lineage selection. If that capacity remains empirically supported and can be grounded in an objective individuating process, Clarke's account can continue to recognize evolutionary individuality even if part of the mechanism lies outside the physical collective.

Bourrat's account deliberately places a stronger condition on that last possibility. It distinguishes collectives whose apparent Darwinian properties arise only from an observer's grouping, collectives whose variation, fitness differences, and heredity are generated by external ecological individuating mechanisms, and collectives in which those properties are maintained through internal individuating mechanisms.<sup>16</sup> Bourrat describes these states as pseudo-Darwinian, Darwinian-like, and Darwinian, respectively. The distinction is intended to capture a transition that can begin through ecological scaffolding but is not completed merely because an external environment reliably organizes particles into higher-level entities. On this account, endogenization matters because the organization supporting higher-level Darwinian properties must become sufficiently internal to provide the collective with resilience to relevant environmental perturbation.<sup>17</sup>

This creates a genuine point of separation from the Clarke account. Bourrat explicitly notes that Clarke's framework can admit external individuating mechanisms when they are sufficiently stable, whereas his own account reserves full Darwinian individuality for cases in which the relevant individuation has become internalized.<sup>18</sup> The disagreement is therefore experimentally consequential. If protected propagule-lineage structure genuinely contributes to higher-level selection but an evolved collective remains dependent on that external relation for the maintenance of its Darwinian organization, the same evidence can support higher-level evolutionary individuality in the Clarke sense while supporting only Darwinian-like status in Bourrat's terminology. No inconsistency follows from those different assessments: they answer different questions about the causal organization of the same collective.

Life-cycle closure introduces a third distinction. Doulcier, Takacs, and Bourrat shift attention from the spatial composition of a candidate individual to the temporal

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<sup>16</sup> Pierrick Bourrat, "Evolutionary Transitions in Individuality by Endogenization of Scaffolded Properties," *The British Journal for the Philosophy of Science* 76, no. 2 (2025): 269–294, <https://doi.org/10.1086/719118>.

<sup>17</sup> Ibid.

<sup>18</sup> Ibid.

organization of a lineage. On their account, a collective-level life cycle exists when an appropriate coarse-graining of lower-level trajectories yields both multiplication and closure.<sup>19</sup> Multiplication requires the production of new candidate collective individuals at the focal level. Closure requires the declared stages to form a connected cycle in which every stage can eventually be reached from every other stage; in the corresponding projection-matrix formulation, this is the condition of irreducibility.<sup>20</sup> When those conditions hold, recurrent stages and multiplication define a collective lineage through time rather than elevating one particular developmental stage to the status of the entire individual.

This temporal criterion is particularly relevant to the WS–SM system. A WS mat alone need not be identified with the individual, nor must the SM phase be treated merely as material exported from an otherwise complete organism. The candidate collective lineage instead extends across the phenotypic transitions through which descendants generate and reestablish the relevant collective stages. The proposed study would therefore ask whether multiplication and closure remain supported by the observed trajectories, including failed transitions and lineage extinction, rather than inferring a life cycle only from lineages that complete it successfully. Experimental handling can provide environmental conditions under which those transitions occur, but moving material between culture vessels cannot by itself count as biological multiplication. The multiplication event must be carried by descendants that establish another instance of the focal collective stage.

The life-cycle account does not require that every causal contribution to this cycle already be internal to the collective. Indeed, ecological scaffolding is one of the mechanisms through which Doulcier and colleagues describe the possible emergence of collective life-cycle closure.<sup>21</sup> This is what makes the account especially informative in the proposed experiment. Relaxing one external scaffold relation could reveal dependence on that relation without necessarily destroying multiplication or closure. A collective lineage could therefore retain a closed life cycle even if another account concludes that an important part of its Darwinian organization remains externally supplied.

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<sup>19</sup> Guilhem Doulcier, Peter Takacs, and Pierrick Bourrat, “Evolutionary Transitions in Individuality and Life Cycle Closure,” *Philosophy of Science* 91, no. 5 (2024): 1393–1403, <https://doi.org/10.1017/psa.2023.162>.

<sup>20</sup> Ibid.

<sup>21</sup> Ibid.

## **What the Post-Evolution Challenge Can Discriminate**

The three accounts overlap on much of the same biology while assigning different significance to particular causal relations. Clarke asks whether objective mechanisms support a capacity for selection at the collective level. Bourrat asks additionally whether the organization responsible for the collective's Darwinian properties remains externally scaffolded or has become internally carried. The life-cycle account asks whether the temporal organization of the lineage supports multiplication and closure at the collective level. The proposed experiment is discriminating because the same scaffold change can bear differently on these properties while the same collective lineage remains the unit of analysis.

One possible outcome is that collective selection capacity and life-cycle closure remain supported while the organization supporting higher-level Darwinian properties proves dependent on external propagule-lineage protection. Other outcomes—including convergence among the accounts, resilience limited to the tested scaffold change, or an unresolved assessment—are equally admissible. The experiment therefore uses the accounts to frame separable empirical questions rather than to privilege a discordant outcome.

## **A Proposed Post-Evolution Scaffold Challenge**

The proposed experiment begins only after a collective-lineage state has evolved under the protected Non-Mixed ecology. This sequencing distinguishes the experiment from the earlier comparison of Non-Mixed and Mixed Propagule treatments, in which lineages experienced their respective metapopulation structures throughout the evolutionary experiment.<sup>22</sup> The present design instead treats the protected ecology as the environment in which the candidate collective organization first develops and the subsequent relaxation of propagule-lineage protection as a challenge to that already evolved state. The experimental question is therefore not which traits evolve under two different environments, but what an evolved collective continues to carry when one relation that previously contributed to its organization is altered.

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<sup>22</sup> Caroline J. Rose, Katrin Hammerschmidt, Yuriy Pichugin, and Paul B. Rainey, “Meta-population Structure and the Evolutionary Transition to Multicellularity,” *Ecology Letters* 23 (2020): 1380–1390, <https://doi.org/10.1111/ele.13570>.

The preferred starting material would be drawn from multiple independently evolved endpoint backgrounds with documented Non-Mixed evolutionary histories. Independence at this stage matters because a result restricted to one unusually successful derived genotype could establish a useful proof of principle while remaining difficult to generalize even within the *Pseudomonas* system. Reference backgrounds should therefore be selected by provenance, viability, and suitability for genealogical reconstruction rather than because previous work suggests that they will produce a desired individuality assessment. Line 17 and its experimentally characterized life-cycle switch remain valuable mechanistic material, but the primary test need not privilege that lineage merely because its transition machinery is already known to be unusually effective.<sup>23</sup> If only a single suitable evolved background were available, the same experiment could still be informative, but its claims would need to be restricted accordingly.

Before scaffold relaxation, the reference state would be characterized prospectively. Previous experiments establish that the system can generate collective-level variation, differential reproductive contribution, lineage-level heredity, recurrent WS–SM transitions, and substantial differences in lineage persistence.<sup>24</sup> Those results make the proposed test feasible; they do not determine in advance that the particular material entering the new experiment possesses all of those properties. The reference characterization would therefore establish the genealogy of the candidate collective lineages, the relevant variation among them, their reproductive contributions across the complete collective cycle, the recurrence of collective traits from parent to descendant lineages, and the distribution of selection between the focal collective level and the constituent-cell level. The same observations would establish the biological stage structure needed to evaluate multiplication and life-cycle closure.

Only after that reference state has been characterized would matched derived material be divided between continued protected propagule-lineage conditions and the relaxed Mixed Propagule condition. The experimental treatment is applied at the metapopulation level: what differs between the two branches is whether propagules from competing collective lineages remain separated during dispersal or enter a shared

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<sup>23</sup> Guilhem Doucier, Philippe Remigi, Daniel Rexin, and Paul B. Rainey, “Evolutionary Dynamics of Nascent Multicellular Lineages,” *Proceedings of the Royal Society B* 292 (2025): 20241195, <https://doi.org/10.1098/rspb.2024.1195>.

<sup>24</sup> *Ibid.*; Caroline J. Rose, Katrin Hammerschmidt, Yuriy Pichugin, and Paul B. Rainey, “Meta-population Structure and the Evolutionary Transition to Multicellularity,” *Ecology Letters* 23 (2020): 1380–1390, <https://doi.org/10.1111/ele.13570>.

competitive pool. Individual lineages, microcosms, generations, and cells therefore provide nested observations within the treated metapopulations rather than independent replicates of the scaffold manipulation. Rose and colleagues' earlier experiments already demonstrate the biological tractability of manipulating this metapopulation relation while retaining the broad WS–SM–WS life-cycle architecture.<sup>25</sup> The proposed experiment changes when that manipulation occurs.

The scaffold contrast should otherwise be kept as narrow as the biological system permits. In particular, the maturation-phase structure, opportunity to generate propagules, single-founder return to the next mat stage, death–birth organization of the experimental metapopulation, and broad phase structure of the life cycle should not be deliberately changed along with propagule mixing. The treatment nevertheless cannot be interpreted as manipulating a single microscopic cause. Pooling propagules simultaneously relaxes lineage assortment and exposes cells from different collective lineages to direct competition. Those effects are biologically linked consequences of changing the protected propagule-lineage relation. The proposed experiment can therefore test whether evolved collective organization depends on that relation as a whole, but it cannot by itself assign any observed dependence uniquely to physical separation or to cross-lineage competition.

This limitation is also why the protected relation should be treated as a candidate individuating mechanism rather than defined as one in advance. For Clarke's account, the relevant question is whether protected propagule-lineage structure actually contributes to the maintenance of heritable differences among collective lineages and thereby to their capacity for selection at the collective level. An experimentally imposed feature is not an individuating mechanism merely because the investigator can manipulate it. Its functional role has to be demonstrated through its consequences for the evolutionary capacity attributed to the collective.

## **Two Timescales of Scaffold Dependence**

A single observation immediately after scaffold relaxation would not be sufficient to characterize what the evolved collective carries. Ecological scaffolding raises a stability problem precisely because a trait that arose under one population structure may

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<sup>25</sup> Caroline J. Rose, Katrin Hammerschmidt, Yuriy Pichugin, and Paul B. Rainey, “Meta-population Structure and the Evolutionary Transition to Multicellularity,” *Ecology Letters* 23 (2020): 1380–1390, <https://doi.org/10.1111/ele.13570>.

respond differently when that structure is changed. Recent theoretical work distinguishes the evolution of a scaffolded trait from its subsequent stability after the scaffolding conditions are perturbed, including the possibility that historical evolution can leave a trait stable even after the environmental conditions responsible for its origin are no longer present.<sup>26</sup> The proposed experiment therefore separates an early response to scaffold relaxation from a longer evolutionary-stability horizon.

The early horizon is intended to characterize what the derived collective brings into the changed environment. If the relevant collective organization deteriorates promptly when propagule-lineage protection is relaxed, that result would support dependence of the antecedent state on the protected ecological relation. It would not show that higher-level organization had never existed; it would show that some part of its maintenance remained contingent on the external condition being tested. Conversely, persistence through the early challenge would show that the organization is not immediately lost when that relation changes, but would not yet establish that the state is evolutionarily stable.

The longer horizon asks whether that persistence itself endures through continued collective generations. This distinction is needed because several trajectories are biologically possible. An evolved collective could retain its organization initially and then erode as selection proceeds in the relaxed environment. It could deteriorate rapidly and later recover through new adaptation. Or it could remain stable across both horizons. These histories should not be treated as equivalent. In particular, recovery after an initial loss would demonstrate evolutionary response to the new environment, not that the state present before the challenge had already been independent of the scaffold. Likewise, persistence across both horizons would demonstrate resilience to the specific propagule-lineage perturbation without establishing independence from every other external relation involved in the collective life cycle.

The exact duration used to distinguish these horizons should not be fixed by conceptual analysis alone. It must be long enough for the early measurement to capture the response of the inherited state and for the later measurement to make evolutionary stability meaningfully answerable in this experimental system.

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<sup>26</sup> Guilhem Doucier, Peter Takacs, Katrin Hammerschmidt, et al., “Stability of Ecologically Scaffolded Traits during Evolutionary Transitions in Individuality,” *Nature Communications* 15 (2024): 6566, <https://doi.org/10.1038/s41467-024-50625-1>.

Determining those intervals, along with the number of independent metapopulations required to distinguish the relevant alternatives, is therefore a matter for specialist experimental design and preregistration rather than a claim of the present proposal.

## **Separating Performance from Heritable Change**

The scaffold challenge also requires a distinction between how a collective performs in the new ecology and what organization its descendants continue to carry. Mixing propagules may immediately reduce collective reproductive success because it alters the competitive environment. Such a result is important, but it does not by itself show that the evolved collective trait has been lost or that the lineage has reverted to an earlier organizational state. A retained trait can perform poorly when expressed under conditions different from those in which it evolved.

For this reason, the proposed study would pair measurements made within the protected and relaxed environments with a standardized common-environment re-assay at both the early and longer horizons. Descendants whose genealogical relationship to the starting lineages is established would be evaluated under the same predefined assay conditions for a set of collective traits specified before the challenge. That set would include properties directly connected to the collective life cycle: reliability of the WS–SM transition, production of descendant mats across the complete collective cycle, recurrence of relevant collective phenotypes, and the heritable contribution of parent lineages to subsequent collective generations. The exact measurement technologies can be refined by specialists, but the distinction between environmental performance and inherited collective organization should be fixed before the outcomes are known.

The resulting comparison allows several otherwise confounded observations to be separated. If descendants perform poorly within the Mixed ecology but retain the relevant collective traits when reassayed under common conditions, the primary effect is consistent with ecological or context-dependent performance. If descendants exposed to the relaxed ecology also exhibit heritable erosion or reorganization when evaluated under the common assay, the evidence for evolutionary instability of the scaffolded collective state becomes substantially stronger. Persistence both within the challenge and after standardized reassay would instead support resilience of the evolved organization to the particular scaffold relation being tested.

The common-environment comparison is not intended to recreate an ideal environment in which individuality is guaranteed to appear. Its role is narrower: the assay must permit descendants of the protected and relaxed branches to be compared directly under the same predefined conditions. The environment and collective traits used for that comparison would therefore need to be specified prospectively, before the treatment histories are interpreted. If the assay itself cannot discriminate retained organization from context-specific expression, the endogenization question would remain unresolved rather than becoming a negative result.

## **Evidence Across the Three Accounts**

The same experimental record can then be evaluated against the questions posed by each account without converting them into a single composite criterion. For Clarke's account, the reference characterization and scaffold challenge would establish whether the candidate collective possesses non-zero capacity for selection at the lineage level and whether that capacity is grounded by an objective individuating mechanism. The protected propagule-lineage relation is the primary candidate demarcating mechanism because it preserves the separation through which heritable differences among collective lineages can matter to their differential contribution. Evidence that relaxing that relation changes the opportunity or response to between-lineage selection would support that functional interpretation. Failure to establish such grounding would weaken the proposed mechanism claim even if group-like phenotypes remained visible.

For Bourrat's account, the reference state must first show that the candidate collective possesses the higher-level variation, fitness differences, heredity, and objective individuation needed for the endogenization question to arise. The scaffold challenge then asks whether the organization supporting those properties remains dependent on the external relation being altered. Demonstrated loss of the relevant collective organization under an adequate challenge would support continued scaffold dependence. Persistence would support independence from that particular relation, but the inference must remain asymmetric: one demonstrated load-bearing external dependency can show that full endogenization has not occurred, whereas survival of

one perturbation cannot show that all relevant external dependencies have disappeared.<sup>27</sup>

The life-cycle assessment uses the same lineages but asks a different empirical question. Biological stages must be defined before the challenge outcomes are interpreted, and the observed trajectory set must include failed transitions, lineage extinctions, mat collapse, and failures to generate the next required stage as well as successful cycles. Recent experiments in this system already show that these lineage histories and failure modes can be reconstructed empirically.<sup>28</sup> The proposed study would use the full trajectory ensemble to ask whether descendant lineages multiply and whether the declared stages remain connected through the recurrent collective cycle. Experimental transfer supplies conditions in which those biological transitions can occur; it does not itself count as the transition. A new culture vessel is not a new collective individual unless biological descendants establish the relevant focal stage.

The three assessments would therefore draw on overlapping evidence while preserving different standards of success. Variation, heredity, reproductive contribution, genealogy, stage transition, and persistence are measured biological properties. Their significance for evolutionary individuality depends on the account asking the question. Preserving that separation is the central methodological commitment of the proposal: a negative result under one account is not imported automatically into another, and inability to measure a required property is not treated as evidence that the property is absent.

The experiment is consequently not designed around a preferred outcome. A lineage may retain higher-level selection capacity while remaining externally dependent; it may retain life-cycle closure while losing another component of Darwinian organization; it may prove resilient to the scaffold challenge; or the proposed distinctions may fail to become empirically resolvable. Each possibility places a different bound on what can be said about the same evolving collective. The value of the experiment lies in making those bounds explicit.

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<sup>27</sup> Pierrick Bourrat, “Evolutionary Transitions in Individuality by Endogenization of Scaffolded Properties,” *The British Journal for the Philosophy of Science* 76, no. 2 (2025): 269–294, <https://doi.org/10.1086/719118>; Guilhem Doulier, Peter Takacs, Katrin Hammerschmidt, et al., “Stability of Ecologically Scaffolded Traits during Evolutionary Transitions in Individuality,” *Nature Communications* 15 (2024): 6566, <https://doi.org/10.1038/s41467-024-50625-1>.

<sup>28</sup> Guilhem Doulier, Philippe Remigi, Daniel Rexin, and Paul B. Rainey, “Evolutionary Dynamics of Nascent Multicellular Lineages,” *Proceedings of the Royal Society B* 292 (2025): 20241195, <https://doi.org/10.1098/rspb.2024.1195>.

## Interpreting the Experimental Outcomes

Interpretation begins with the reference state rather than with the scaffold perturbation. The endogenization question is meaningful only if the candidate collective first displays the higher-level properties whose causal support is being tested. If the reference populations do not exhibit reproducible collective variation, differential reproductive contribution, heredity, and an empirically supported higher-level selection capacity, then the proposed experiment has not uncovered scaffold-dependent individuality; it has failed to establish the antecedent collective organization required for that question. Likewise, if the observed trajectories do not support multiplication and closure at the collective-lineage level, life-cycle closure is not supported regardless of what happens after propagule mixing. These are biologically informative outcomes rather than failed experiments, because they locate the point at which the proposed higher-level interpretation ceases to be supported.

The result of greatest discriminatory interest would arise if the reference state supports focal-level selection capacity and collective life-cycle closure, while relaxation of propagule-lineage protection reveals that an important part of the Darwinian organization remains externally dependent. Under those conditions, Clarke's forwards-looking account could continue to recognize the candidate as an evolutionary individual because an objective individuating mechanism supports selection at the collective level. Bourrat's account could classify the same collective as Darwinian-like rather than fully Darwinian if the experiment shows that a load-bearing component of its individuation remains environmentally supplied. The life-cycle account could simultaneously continue to support life-cycle closure if multiplication and closure continue to characterize the lineage.<sup>29</sup> The significance of such a result would lie precisely in the fact that no biological object has changed between the assessments. The disagreement would arise from different account commitments applied to the same experimentally characterized lineage.

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<sup>29</sup> Ellen Clarke, "A Levels-of-Selection Approach to Evolutionary Individuality," *Biology & Philosophy* 31 (2016): 893–911, <https://doi.org/10.1007/s10539-016-9540-4>; "Evolutionary Individuals," in *The Units of Life: Kinds of Individual in Biology* (Oxford University Press, 2025), 81–120, <https://doi.org/10.1093/9780191948008.003.0005>; Pierrick Bourrat, "Evolutionary Transitions in Individuality by Endogenization of Scaffolded Properties," *The British Journal for the Philosophy of Science* 76, no. 2 (2025): 269–294, <https://doi.org/10.1086/719118>; Guilhem Doulier, Peter Takacs, and Pierrick Bourrat, "Evolutionary Transitions in Individuality and Life Cycle Closure," *Philosophy of Science* 91, no. 5 (2024): 1393–1403, <https://doi.org/10.1017/psa.2023.162>.

That outcome is only one branch of the experiment. The protected propagule-lineage relation may fail to behave as the proposed demarcating mechanism. Relaxing it could leave between-lineage selection capacity essentially unchanged, or the reference state itself may prove insufficient to establish a meaningful higher-level selection capacity. In either case, the Clarke-based interpretation would need to narrow accordingly. Alternatively, scaffold relaxation could disrupt the biological transitions needed for collective multiplication or prevent reestablishment of the declared stages. That result would place pressure on the life-cycle assessment independently of whether higher-level variation or heredity remained detectable. The experiment therefore permits the accounts to converge as well as diverge.

Persistence under scaffold relaxation must remain bounded in interpretation. Stability through both the early and longer horizons would support resilience to the particular protected propagule relation being tested, not complete endogenization while other external contributions remain untested.<sup>30</sup> Conversely, an immediate performance decline without heritable erosion in the common-environment re-assay would indicate environmental sensitivity rather than loss of internally carried organization, while early deterioration followed by later recovery would indicate adaptation to the changed environment rather than antecedent independence from the scaffold.<sup>31</sup>

## **Conditions That Would Rule Out the Discordant Outcome**

Because the experiment is intended to discriminate among accounts rather than to illustrate a preferred philosophical conclusion, the central discordance result must be capable of failing for substantive reasons. The most direct failure would occur if the reference state does not establish the higher-level properties needed to pose the comparison. Without supported collective variation, reproductive differences, heredity, or focal-level selection capacity, there is no basis for treating the material as an evolutionary individual under Clarke's account or as Darwinian-like under Bourrat's account. Prior work can motivate the experiment and establish that such properties are possible in the system, but it cannot substitute for their prospective demonstration in the reference populations used in the new study.

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<sup>30</sup> Pierrick Bourrat, "Evolutionary Transitions in Individuality by Endogenization of Scaffolded Properties," *The British Journal for the Philosophy of Science* 76, no. 2 (2025): 269–294, <https://doi.org/10.1086/719118>; Guilhem Doucier, Peter Takacs, Katrin Hammerschmidt, et al., "Stability of Ecologically Scaffolded Traits during Evolutionary Transitions in Individuality," *Nature Communications* 15 (2024): 6566, <https://doi.org/10.1038/s41467-024-50625-1>.

<sup>31</sup> Ibid.

The Clarke-based interpretation would also fail if protected propagule-lineage structure cannot be shown to contribute functionally to the relevant higher-level selection capacity. A treatment effect is not enough merely because the manipulated feature appears biologically important. If propagule mixing changes collective performance without altering the heritable between-lineage differentiation or selection opportunity that the demarcation claim is meant to explain, the proposed identification of that relation as a demarcating mechanism would need to be withdrawn or narrowed. The experiment would still yield information about the ecology of the collective, but not the particular individuating function proposed here.

The inference under Bourrat's account would fail for a different reason if apparent scaffold dependence reduced only immediate performance while the collective organization remained heritably intact. In that case, the experiment would have shown environmental sensitivity rather than loss of internally carried organization. Conversely, if the lineage remains stable after the perturbation, the hypothesis that this specific external relation remains necessary for maintaining the relevant collective organization would not be supported. The result would then shift from demonstrated dependence to bounded resilience, leaving the broader endogenization question open rather than resolving it in the opposite direction.<sup>32</sup>

The life-cycle interpretation can fail independently. If the prospectively declared stage structure does not yield biological multiplication, if descendant lineages do not reestablish the focal stages, or if the trajectory graph fails the closure condition, the proposed collective life cycle is not supported under that operationalization.<sup>33</sup> Such a result cannot be repaired by redefining stages after the outcome is known or by restricting analysis to the surviving lineages that happened to complete the cycle. Failed transitions and extinction must remain in the analysis because they help determine whether the proposed life-cycle organization is actually characteristic of the candidate lineage population.

Some failures would prevent a clear inference rather than provide evidence against the relevant biological property. If lineage ancestry cannot be resolved after propagule mixing, the experiment may be unable to assign reproductive contribution or heredity at the focal level. If the scaffold manipulation also changes neighboring aspects of the

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<sup>32</sup> Ibid.

<sup>33</sup> Guilhem Doucier, Peter Takacs, and Pierrick Bourrat, "Evolutionary Transitions in Individuality and Life Cycle Closure," *Philosophy of Science* 91, no. 5 (2024): 1393–1403, <https://doi.org/10.1017/psa.2023.162>.

life cycle so extensively that protected lineage structure cannot be isolated even at the intended relational level, causal interpretation may become impossible. If the common-environment assay cannot distinguish inherited organization from environment-specific expression, the endogenization question remains unresolved. In each case, lack of an answer should remain distinct from evidence for a negative answer.

## Scope of the Proposed Test

Even under ideal execution, this first experiment would answer a deliberately bounded question. It tests dependence on one identified ecological relation in one experimentally tractable system. The Non-Mixed-to-Mixed transition does not remove every external contribution to the collective life cycle, and the experiment therefore cannot establish complete autonomy or full endogenization simply because the candidate survives that perturbation. Other features of the experimental ecology—including the structure of collective maturation, opportunities for propagation, single-founder establishment, and the externally organized sequence of life-cycle phases—may continue to perform causal work. The first study is designed to identify what follows from relaxing one especially informative relation, not to erase the entire experimental environment.

The same restraint applies to generalization beyond the *Pseudomonas* system. These lineages are valuable precisely because their collective stages, genealogy, competing levels of selection, and environmental scaffolding can be manipulated with unusual precision. Those advantages do not make them representative of every evolutionary transition in individuality. A result reproduced across several independently evolved endpoint backgrounds would support generality within this experimental family more strongly than a result obtained from one genotype, but even that would not warrant a universal claim about multicellularity, symbiosis, eusociality, or other major transitions.

The proposal also leaves several execution choices intentionally unresolved. Exact replication, the duration of the early and longer horizons, lineage-marking technology, the statistical treatment of hierarchical and repeated observations, and the operational definition of a successful common-environment comparison require specialist judgment grounded in the actual biological material and measurement system. Fixing those parameters conceptually before such consultation would create an appearance

of precision without improving the scientific test. What must remain fixed are the empirical requirements those methods have to satisfy: ancestry must be resolvable, collective and lower-level contributions must be distinguishable, life-cycle stages must be prospectively defined, and retention claims must be supported affirmatively rather than inferred from a failure to detect change.

These limits do not diminish the purpose of the experiment. They define it. Evolutionary individuality encompasses several biological questions that often coincide in mature organisms but can separate during evolutionary transitions. The proposed scaffold challenge exploits that separation rather than attempting to eliminate it. Its contribution would be to evaluate higher-level selection capacity, endogenization, and life-cycle closure in the same experimental system and ask how far the evidence supports each of them.

If the three accounts converge, the experiment would show that their different criteria happen to align in the tested lineage under the tested conditions. If they diverge, the result would identify which biological commitment produces that separation. If the evidence proves inadequate to decide among them, the source of uncertainty would itself become explicit. In none of these cases is the experiment required to select a single definition of evolutionary individuality. Its narrower aim is to turn conceptual differences into questions that experimental evidence can answer, constrain, or leave open.

## **Discussion**

Evolutionary individuality is often difficult to test because many of its proposed features coincide in familiar organisms. A mature multicellular organism can possess strong parent–offspring continuity, mechanisms that suppress conflict among parts, reproductive specialization, a closed developmental cycle, and a clear capacity for selection at the organismal level. In such cases, different accounts of individuality may reach the same conclusion even when they assign explanatory importance to different biological properties. Transitional systems are more revealing because those properties can separate. Ecological scaffolding makes that separation especially visible:

higher-level evolutionary organization can arise while some of the causal work supporting it remains outside the collective itself.<sup>34</sup>

This approach also changes what should count as an informative experimental result. A clean divergence among the accounts would be valuable because it would identify a biological condition under which their commitments come apart. But convergence would be equally informative. If scaffold relaxation simultaneously undermines higher-level selection capacity, endogenized organization, and collective life-cycle closure, then several theoretically distinct criteria would have converged on the same biological transition in this system. If all three remain supported, the experiment would instead identify a lineage in which multiple dimensions of individuality persist despite the tested ecological change. The experiment therefore does not depend on discordance for its scientific value.

Indeterminate outcomes matter for the same reason. Some questions about individuality can fail to become answerable because the underlying biological relations cannot be measured cleanly. Loss of lineage resolution after mixing, inability to distinguish environmental performance from inherited organization, or failure to define stable biological stages would limit what can be inferred without showing that the relevant property is absent. Treating such cases explicitly as unresolved prevents methodological failure from being converted into a biological conclusion. It also makes clear which additional measurements or experimental refinements would be required before the individuality question could be reopened.

The post-evolution ordering of the scaffold challenge is particularly important in this respect. Earlier experiments established that different metapopulation structures can direct the evolution of different collective properties.<sup>35</sup> The proposed study asks a different question: once a collective organization has arisen, how much of that organization persists when one relation that helped structure its evolutionary environment is changed? This shift from origin to persistence makes it possible to distinguish the conditions under which a higher-level trait evolved from the conditions required for its continued maintenance. Theoretical work on ecological scaffolding

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<sup>34</sup> Andrew J. Black, Pierrick Bourrat, and Paul B. Rainey, “Ecological Scaffolding and the Evolution of Individuality,” *Nature Ecology & Evolution* 4 (2020): 426–436, <https://doi.org/10.1038/s41559-019-1086-9>; Pierrick Bourrat, “Evolutionary Transitions in Individuality by Endogenization of Scaffolded Properties,” *The British Journal for the Philosophy of Science* 76, no. 2 (2025): 269–294, <https://doi.org/10.1086/719118>.

<sup>35</sup> Caroline J. Rose, Katrin Hammerschmidt, Yuriy Pichugin, and Paul B. Rainey, “Meta-population Structure and the Evolutionary Transition to Multicellularity,” *Ecology Letters* 23 (2020): 1380–1390, <https://doi.org/10.1111/ele.13570>.

and trait stability already identifies that distinction as central to endogenization.<sup>36</sup> The proposed experiment brings it into a biological system in which collective genealogy, reproductive success, stage transitions, and lineage failure can all be observed directly.

The design therefore offers a way to examine evolutionary transitions without requiring that individuality appear all at once. A transition may involve the sequential acquisition, stabilization, or internalization of several features that later become tightly associated. Selection at the collective level may become consequential before all supporting organization is internally carried. A reproducible life cycle may become established while some of the ecological relations that sustain it remain externally imposed. Internal mechanisms may later reduce dependence on those relations without creating an entirely new biological entity at a single threshold. The proposed experiment is compatible with such partial and staged transitions because it asks which capacities are supported at each point rather than presuming that every relevant feature must cross the same boundary simultaneously.

The broader value of the proposal is therefore methodological as much as system-specific. Debates about biological individuality often become difficult to compare experimentally when different accounts are applied to different organisms, different levels, or differently described biological processes. The proposed design holds those variables as stable as the system allows. The candidate remains the same collective lineage, the experimental evidence comes from the same metapopulation experiment, and the accounts are evaluated without changing their constitutive criteria after the result is known. This does not eliminate philosophical disagreement, but it narrows the disagreement to identifiable biological commitments that can be investigated experimentally.

The same strategy may eventually be applicable beyond this particular experimental family. Other transitions—in multicellular, symbiotic, microbial, or social systems—may also contain cases in which selection capacity, reproductive organization, and internalization of individuating mechanisms do not appear together. Whether the specific scaffold manipulation proposed here generalizes is an empirical question, not an assumption. What can generalize more readily is the experimental discipline: identify the candidate individual independently, keep the biological target

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<sup>36</sup> Guilhem Doucier, Peter Takacs, Katrin Hammerschmidt, et al., “Stability of Ecologically Scaffolded Traits during Evolutionary Transitions in Individuality,” *Nature Communications* 15 (2024): 6566, <https://doi.org/10.1038/s41467-024-50625-1>.

fixed, determine which causal relations each account treats as constitutive, and design interventions capable of changing those relations without silently changing the object under study.

## Conclusion

The proposed experiment treats evolutionary individuality as something that can be investigated through separable biological commitments rather than settled by choosing a definition in advance. Experimentally evolved *Pseudomonas fluorescens* lineages provide a tractable system because collective reproduction, genealogy, selection at multiple levels, and ecological scaffolding can all be observed and manipulated within the same lineage architecture. Earlier experiments established how protected and mixed propagule ecologies alter evolutionary response; the proposed study extends that work by asking what an already evolved collective continues to carry after protected propagule-lineage structure is relaxed.<sup>37</sup>

The answer need not be singular. The same lineage may retain a capacity for higher-level selection, remain dependent on an external individuating relation, and continue to satisfy collective life-cycle closure. It may lose several of those properties together. It may prove resilient to the tested scaffold change, or the available evidence may leave one or more questions unresolved. Each result constrains a different claim about evolutionary individuality.

The experiment is intentionally narrower than a general theory of evolutionary transitions. It tests one scaffold relation in one experimentally developed system and preserves several execution decisions for specialist design. Its value lies in making a difficult conceptual disagreement open to direct empirical test. By holding the lineage under study fixed while allowing the accounts to retain their own criteria, the proposed study asks not which definition of individuality should prevail, but what the evolving collective itself permits us to say.

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<sup>37</sup> Caroline J. Rose, Katrin Hammerschmidt, Yuriy Pichugin, and Paul B. Rainey, “Meta-population Structure and the Evolutionary Transition to Multicellularity,” *Ecology Letters* 23 (2020): 1380–1390, <https://doi.org/10.1111/ele.13570>.

## **AI Use Disclosure**

Artificial intelligence tools were used during the development of this work for research assistance, adversarial review, conceptual organization, drafting support, editing, and formalization support. The author directed the research program, selected and evaluated the scientific claims and sources, determined the experimental proposal and its scope, and takes responsibility for the final content.

# Evolutionary Individuality as an Empirical Question

*A Proposed Experimental Test in Pseudomonas*

Signed,  
  
Rick